

Contest Log Analytics - User Guide

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--- Revision History ---

[1.0.0-alpha.18] - 2026-01-31

Changed

- Updated version to match project release v1.0.0-alpha.18 (no content changes)

[1.0.0-alpha.17] - 2026-01-30

Added

- Report List (pull-out drawer) on main, QSO, and multiplier dashboards; hierarchical list of generated reports (same as ZIP); filename-based labels; search, collapse, active-report highlight.

Changed

- Updated version to match project release v1.0.0-alpha.17

[1.0.0-alpha.16] - 2026-01-30

Changed

- Updated version to match project release v1.0.0-alpha.16 (no content changes)

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Changed

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Changed

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Changed

- Updated version to match project release v1.0.0-alpha.11 (no content changes)

[1.0.0-alpha.10] - 2026-01-24

Changed

- Updated version to match project release v1.0.0-alpha.10 (no content changes)

[1.2.0] - 2025-12-23

Added

- Added Multiplier Breakdown reports (HTML, Text, JSON) to Section 5.

[1.1.0] - 2025-12-15

Added

- Added Section 6 "The Web Dashboard" to document the containerized workflow.

[1.0.0] - 2025-12-06

Added

- Added `wrtc_propagation` and `wrtc_propagation_animation` to the list

of Available Reports.

[0.94.0-Beta] - 2025-12-06

Removed

- Removed `multipliers_by_hour` from the list of Available Reports.

[0.91.2-Beta] - 2025-10-11

Changed

- Updated the "Supported Contests" list to be a more user-friendly

conceptual list, rather than a list of specific CONTEST: tags.

- Added the missing `--wrtc` command-line option.

[0.90.16-Beta] - 2025-10-06

Added

- Added the complete list of valid CONTEST: tags to Section 4.

- Added a new subsection to Section 3 explaining the `<EventID>`

component of the output³ directory structure.

[0.90.15-Beta] - 2025-10-06

Fixed

- Added the missing `CQ160mults.dat` file to the list of required

1. Introduction: What is the Contest Log Analytics?

The Contest Log Analytics is a web-based analysis tool designed for amateur radio contesters who want to perform deep, data-driven analysis of their operating performance. It goes beyond the simple score summary provided by most logging software, allowing you to:

- Process raw Cabrillo log files into a clean, standardized format.
 - Automatically classify every QSO as "Run," "Search & Pounce," or "Unknown" to analyze your operating strategy.
 - Generate detailed reports and charts that compare your log against one or more others.
 - Analyze performance on a band-by-band basis to identify strengths and weaknesses.
 - Calculate contest-specific QSO points for supported contests. The ultimate goal of this program is to help you understand your contest operation in minute detail, identify missed opportunities, and improve your strategy for the next event.
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2. What You Need to Get Started

Before running the analyzer, you will need:

- **Your Cabrillo Log File(s):** These are the standard log files generated by your contest logging software (e.g., `kd4d.log`, `n0ni.log`). You can analyze a single log or compare multiple logs at once (up to 3 logs).
- **Docker:** The application runs in a Docker container. See the Installation Guide for setup instructions.

The analyzer automatically detects the contest type from the `CONTEST:` field in your Cabrillo file header and applies the correct rules. All required data files (`CTY.DAT`, multiplier files, etc.) are included in the Docker container.

3. The Web Dashboard

The Contest Log Analytics features a web-based interface designed for ease of use.

Launching the App

1. Ensure Docker is running.
2. Navigate to the project root.
3. Run: `docker-compose up --build`
4. Open your browser to <http://localhost:8000> (development) or visit <https://cla.kd4d.org> (production).

Upload Options

The web dashboard supports flexible log analysis:

- **Single-Log Analysis:** Upload only Log 1 to analyze your performance, rates, and multipliers in detail.
- **Comparison Analysis (2-3 Logs):** Upload 2 or 3 logs to enable head-to-head comparison with unique vs. common QSO breakdowns.
- **Public Archive Integration:** For supported contests (CQ WW, CQ 160, ARRL 10, ARRL DX, IARU HF), you can fetch competitor logs directly from public archives without manual downloads.

Identity Agnostic Workflow

The analyzer does not know "who you are." It simply compares logs.

- **Log 1:** Upload your primary log here (The "Hero"). This slot is required.
- **Log 2 & 3:** Upload competitor logs here (The "Villains"). These slots are optional.

The Dashboard

Once analyzed, you are presented with a "Strategy Board":

- **The Scoreboard:** High-level metrics including Run % and total score.
- **The Triptych:** Three paths for deeper analysis (Animation, QSO Reports, Multiplier Reports).
- **Report List (Drawer):** A pull-out arrow on the left edge (halfway down) opens a list of all generated reports in the same hierarchy as the ZIP. Click any report to open it in the full-screen overlay. The list uses actual filenames (callsign and band when present); search and category collapse are available.
- **Raw Data:** A button to download the full set of generated reports as a ZIP file.

For Single-Band, Multi-Mode Contests: For contests like ARRL 10 Meter that operate on a single band with multiple modes, the QSO dashboard shows "Mode Activity" instead of "Band Activity", and the Multiplier dashboard breaks down multipliers by mode (CW, SSB, etc.) rather than by band.

Sweepstakes-Specific Features

ARRL Sweepstakes includes specialized analysis features in the Multiplier Dashboard:

- **Fixed Multiplier Scale:** Progress bars use a fixed scale showing 85 sections (the maximum possible). A vertical reference line at 100% indicates the maximum multiplier count.
- **Multiplier Saturation Visualization:** When all logs have reached 85 multipliers, the dashboard clearly shows this saturation state.

- **Enhanced Missed Multipliers Report:** A detailed breakdown showing which logs worked each missed section, bands/modes where worked, and Run/S&P/Unknown counts. This report appears in the "Missed Multipliers" tab alongside the standard missed multipliers report.
- **Band Spectrum Suppression:** For Sweepstakes, the Band Spectrum visualization is automatically suppressed since multipliers count once per contest (not per band), making per-band breakdowns less meaningful.

WRTC Scoring

For IARU-HF logs, you can apply WRTC scoring rules by selecting "Apply WRTC scoring rules" in the advanced options and choosing the WRTC contest year (e.g., WRTC-2026).

4. Supported Contests

The analyzer uses the **CONTEST:** field in your Cabrillo file header to automatically apply the correct rules. The following contests are currently supported:

- ARRL 10 Meter
- ARRL DX (CW & SSB)
- ARRL Field Day
- ARRL Sweepstakes
- CQ 160-Meter
- CQ WPX (CW & SSB)
- CQ World Wide DX (CW & SSB & RTTY)
- IARU HF World Championship
- North American QSO Party (NAQP)
- WAE (CW & SSB)
- WRTC (via IARU-HF logs with WRTC scoring option)

5. Available Reports

All reports are automatically generated and available for viewing in the web dashboard or download as a ZIP archive.

Animation Reports (animations/)

- `hourly_animation`: Hourly Rate Animation
- `wrtc_propagation_animation`: WRTC Propagation Animation

HTML Reports (html/)

- `html_multiplier_breakdown`: Multiplier Breakdown Report (Group Par)

- `html_qso_comparison`: HTML QSO Comparison Report

Chart Reports (`charts/`)

- `chart_point_contribution`: Point Contribution Breakdown (Single Log)
- `qso_breakdown_chart`: QSO Breakdown Chart

Plot Reports (`plots/`)

- `band_activity_heatmap`: Band Activity Heatmap
- `comparative_band_activity`: Comparative Band Activity (shows Mode Activity for single-band, multi-mode contests like ARRL 10 Meter)
- `comparative_band_activity_heatmap`: Comparative Band Activity Heatmap
- `comparative_run_sp_timeline`: Comparative Activity Timeline (Run/S&P)
- `cumulative_difference_plots`: Cumulative Difference Plot
- `point_rate_plots`: Cumulative Point Rate Plot
- `qso_rate_plots`: Cumulative QSO Rate Plot
- `wrtc_propagation`: WRTC Propagation by Continent

Text Reports (`text/`)

- `comparative_continent_summary`: Comparative Continent QSO Summary
- `text_multiplier_breakdown`: Multiplier Breakdown (Group Par)
- `rate_sheet_comparison`: Comparative Rate Sheet
- `comparative_score_report`: Comparative Score Report
- `continent_breakdown`: Continent Breakdown by Run/S&P
- `continent_summary`: Continent Summary
- `missed_multipliers`: Missed Multipliers
- `multiplier_summary`: Multiplier Summary
- `json_multiplier_breakdown`: JSON Multiplier Breakdown Artifact (Data Source)
- `qso_comparison`: QSO Comparison Summary
- `rate_sheet`: Hourly Rate Sheet (per hour)
- `score_report`: Score Report
- `summary`: QSO Summary by Run/S&P
- `text_wae_comparative_score_report`: WAE Comparative Score Report
- `text_wae_score_report`: WAE Score Summary

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6. Report Interpretation

Reports are organized into several categories:

Single vs. Multi-Log Features

- **Single-Log Reports:** Score summaries, rate sheets, hourly breakdowns, multiplier analysis, and performance metrics.
- **Multi-Log Reports:** All single-log features plus cumulative difference plots, QSO comparison charts, missed multiplier analysis, and head-to-head rate comparisons.